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Paradigm Shift in Well Intervention Using Advanced Wireline Tractor for Challenging Horizontal Completions

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Abstract

Horizontal well completions are widely recognized as the most cost-effective method to enhance reservoir production by increasing the contact area between the wellbore and formation. However, rig-less access for reservoir surveillance and intervention in these wells has been a persistent challenge. In addition, conveyance in extended reach and maximum reservoir contact (MRC) wells with single or multilateral horizontal, and more than five Km of total reservoir contact, obtaining downhole data requires extensive simulation with accurate friction factors.

Historically, Coil Tubing (CT) has been the preferred method for intervening in horizontal wells, as most existing wireline tractors were limited to cased-hole wells due to the complex geometry of open-hole wellbores, the presence of restrictions, washouts, and rock debris. CT, however, has limitations, such as the buckling effect, which reduces the depth reach in extended-reach wells and chokes well flow during logging operations, adversely impacting the quality of data acquisition. Additionally, CT requires a larger footprint, more logistics at the well site, and longer operating times during logging compared to wireline tractors.

This study highlights a paradigm shift in well intervention with successful deployment of a new generation of slimhole wireline tractors in oil and gas wells, marking the transition from Coil Tubing (CT) to more efficient wireline technology. These advanced tractors are designed to convey a range of tools, including production and reservoir monitoring tools, as well as mechanical intervention tools, to diagnose and intervene in challenging horizontal completions and extended-reach wells, ultimately improving well productivity. Several key improvements over conventional tractors have been incorporated, such as enhanced gripping capabilities, reversed tractoring, increased debris tolerance, H₂S resistance, the ability to log while tractoring, and extended reach capabilities that surpass those of CT. These innovations have not only enhanced operational efficiency but also reduced intervention time and equipment footprint, contributing to more sustainable operations by cutting CO₂ emissions by over 70% compared to prolonged well interventions using CT.

This study highlights the benefits of adopting the new tractor system for well interventions, offering enhanced operational efficiency, reduced environmental impact, and optimized reservoir characterization. By improving intervention time and reducing CO₂ emissions with substantial ground water conservation,

this solution not only enhances reservoir understanding but also supports the sustainability goals of future oil and gas operations at a lower cost.

Introduction

Effective reservoir monitoring is a critical aspect of maximizing hydrocarbon recovery and ensuring the long-term productivity of oil and gas wells. Throughout the life of a well, various forms of well intervention are required to gather downhole data and perform necessary treatments. Among these, achieving maximum reach in horizontal wells, particularly in a cost-effective manner, remains a key objective for production engineers. Interventions such as production logging and water shut-off operations demand reliable conveyance methods capable of accessing extended lateral sections of the wellbore.

To address these challenges, a new generation wireline tractor has been deployed in multiple openhole horizontal producers with diverse geometries. This advanced conveyance technology enabled the successful delivery of Multiphase Production Logging Tools (MPLT) and Pulsed Neutron Logging (PNL) tools to the required target depths. Compared to traditional coiled tubing techniques, the new wireline tractor demonstrated superior reach and operational efficiency. The enhanced performance, evident in both upward and downward logging passes, resulted in a twofold increase in conveyance effectiveness relative to conventional tractor systems. This breakthrough represents a significant advancement in well diagnostics and intervention capabilities, ultimately supporting better reservoir management and production optimization.

This paper discusses the challenges associated with logging and conveyance operations in complex openhole and smart casedhole horizontal wells. It explores the selection of optimal logging tools and conveyance strategies aimed at enhancing operational efficiency, improving data quality, and ensuring reliable data acquisition. By addressing these key aspects, the paper highlights how advanced conveyance solutions contribute to more effective reservoir monitoring and improved reservoir management decisions.

Completion Overview and Conveyance Challenges

The wells discussed in this paper were drilled and completed as horizontal barefoot completions. Each well features a 7" liner extending to the start of the horizontal section, followed by a 6" to 6 ½" openhole lateral. The average length of the horizontal reservoir drain is approximately 3,500 to 4,500 ft per well. The upper completion consists of 3 ½" to 4 ½" production tubing, as illustrated in Fig. 1.

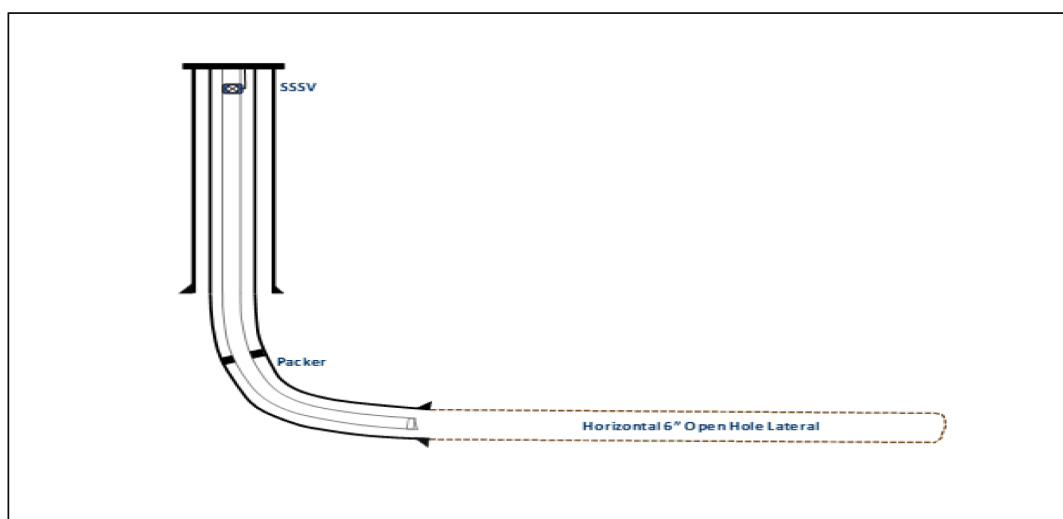


Figure 1—Typical Well Completion

Multiphase Production Logging (MPLT) and Pulsed Neutron (PNL) logs in openhole horizontal wells is essential for understanding reservoir performance, optimizing hydrocarbon recovery, and diagnosing operational issues. However, compared to vertical wells, horizontal openhole completions present unique challenges that complicate both data acquisition and interpretation.

One major issue is the behavior of multiphase flow in a horizontal wellbore. While gravity segregation is relatively minor in vertical wells, it becomes significant in horizontal sections. Gas tends to migrate to the top of the wellbore, water settles at the bottom, and oil occupies the space in between. This fluid segregation leads to complex flow regimes, such as slug, annular, or stratified flow, which can distort measurements from traditional logging tools. These tools often record only the flow in their immediate vicinity, missing contributions from other fluid phases unless specially designed for horizontal or multiphase environments.

Another critical challenge is tool conveyance. Reaching total depth (TD) in long horizontal laterals is often limited by drag and friction. Coiled tubing (CT), commonly used for deployment, may encounter helical lockup, a condition where the coil tubing coils within the wellbore and can no longer advance, limiting access to the toe. Mitigating this requires friction reducers, which can negatively impact sensor readings, particularly spinners and holdup probes, or using larger CT diameters to increase reach. In addition, excessive use of the friction reducers is detrimental to the reservoir connectivity to the wellbore, leading to reduced well productivity.

Even when TD is reached, larger-diameter CT can restrict wellbore flow, especially in wells with smaller production tubing. This choking effect alters flow conditions during logging, affecting measurements of phase distribution and zone contributions. Additionally, during the upward logging pass, CT movement can swab the well, unintentionally drawing in fluids and further distorting flow profiles.

These operational and flow related complications make accurate production logging in openhole horizontal wells particularly difficult. Inaccurate quantification of oil and water contributions can compromise water shut-off decisions and lead to flawed reservoir models, impacting long-term well and field performance.

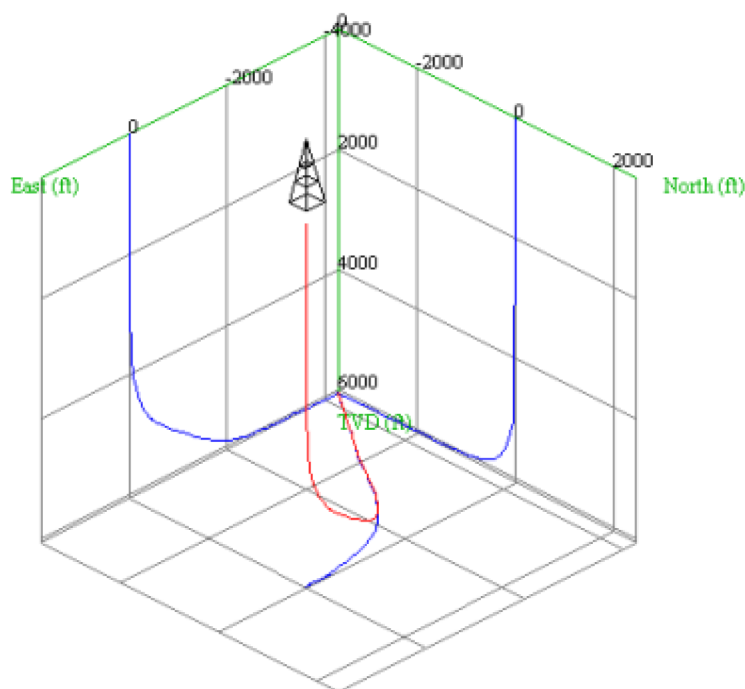


Figure 2—Typical Wellbore Trajectory

Slim Openhole Tractor

The Slim Open Hole tractor works on an inch worm mechanism made possible by 2 different reciprocating arms which are moving in tandem. The arm design (Fig. 3) provides a large grip area reducing the stress on the formation enabling the tool to operate in extremely low formation compressive strengths. This design is an improvement over the existing wheel-based tractor solutions as the wheel-based systems provide a point contact and hence increases stress on formation.

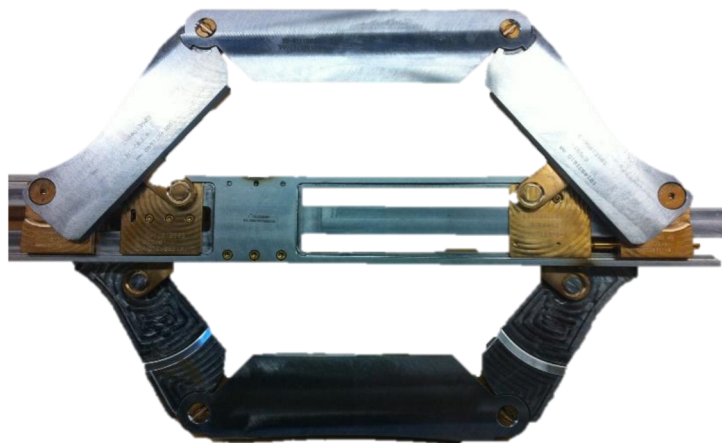


Figure 3—Grip Mechanism of Tractor

The tractor's small outer diameter of 2 1/8" and high expansion ratio allow it to access boreholes up to 10 1/2" in diameter. This enables the tractor to pass through narrow production tubing and reach larger borehole sizes efficiently.

It features automated arm adjustment, requiring minimal user input during tractoring. This automation allows the tractor to navigate washouts and borehole restrictions effectively, an essential capability for barefoot completions, where the tool must adapt to rugose and non-circular borehole geometries.

Its consistent tractoring speed and the ability to collect data while moving downhole contribute to high-quality production logging.

The components exposed to wellbore fluids are constructed from materials compatible with H₂S and CO₂ environments. The tractor employs Inchworm traction for continuous motion, is suitable for formations with low unconfined compressive strength, and provides real-time surface readout while tractoring.

The scissor-style closing mechanism of the tractor arms minimizes debris accumulation and ensures safe tool retrieval to surface after logging. Controlled tractoring forces reduce the risk of damage to slim production logging tools.

Operating at over 40% efficiency, the system optimizes surface power usage, outperforming conventional tractor technologies. It is capable of deployment on various cable types to depths beyond 40,000 feet. Proven wireline pressure control equipment addresses health, safety, and environmental concerns, while also streamlining procedures and reducing operational time and manpower.

The tractor is engineered for use in both open-hole and cased-hole applications across oil, water, and gas wells.

Multiphase Production Logging Tool (MPLT)

In a multiphase flow system, buoyancy due to well deviation causes the fluid to separate into different phases with a mixing layer in between. Gravity ensures that the light phase fluid travels at a faster speed than the heavier phase fluid, with deviation less than 90 deg and vice versa in wells with deviation above 90 deg.

Biphasic experiments carried out in a controlled flow loop with equal flow rates for oil and water show the dramatic effects of borehole deviation on flow behavior (Fig. 4). At 90 deg deviation, the velocities and holdups of oil and water are nearly equal. At deviation lower than 90 deg (uphill), water (heavier phase) slows down, and the oil velocity increases as the water holdup increases while the oil holdup decreases. At well deviation above 90 deg (downhill), flow is still predominantly stratified. The water flows much faster than the oil because of its higher fluid density as the water holdup now decreases while the oil holdup increases.

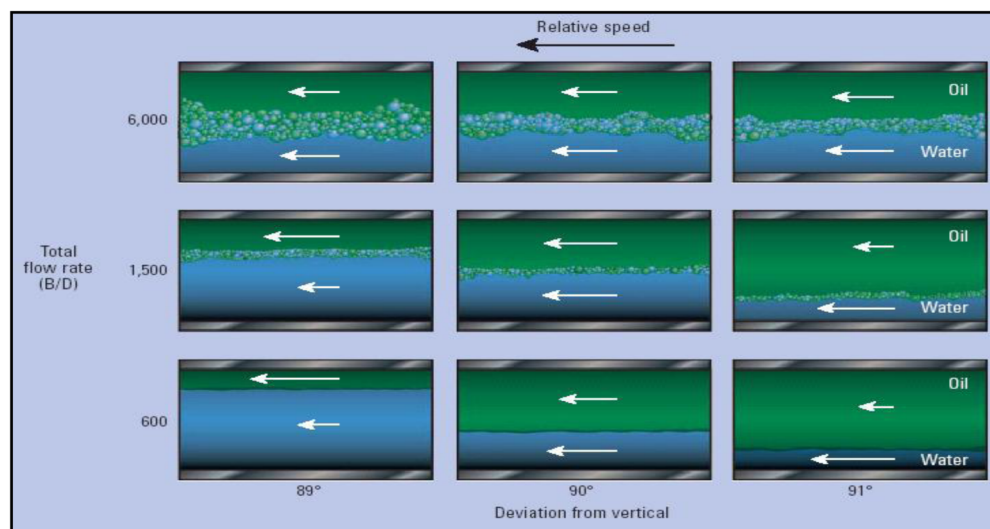


Figure 4—Flow loop experiment for oil and water flow.

In these complex conditions, the multiphase production logging tool (MPLT), shown in Fig. 5, is used to measure the velocity profile along the transverse diameter of the wellbore. It provides measurements of mixed and segregated flow regimes and detects downhole water recirculation (Baldauff et al. 2004). This MPLT incorporates 5 spinners to measure the velocity distribution and a pair of 6 Electrical and optical probes to measure respectively water and gas holdups across the vertical axes of the wellbore cross-section.

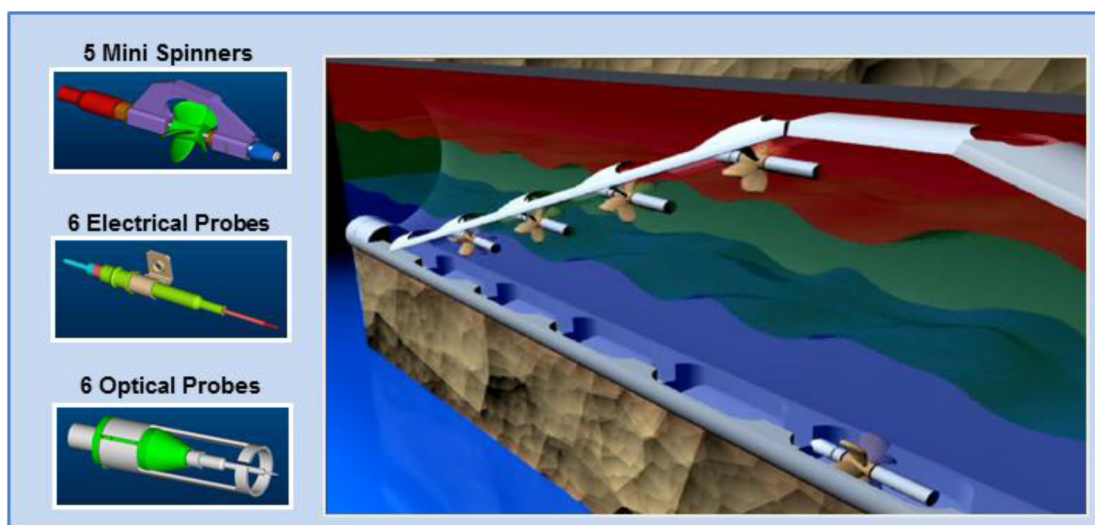


Figure 5—Multi-array production logging tool location of multiple probes and spinners.

The tool independently measures the velocities of water and oil in complex stratified horizontal wells to provide an accurate oil and water flow profile.

New Generation Slim Pulsed (PNL)

The New generation Pulsed Neutron (Fig. 6) primary mission is formation evaluation and reservoir monitoring in cased boreholes for both horizontal and vertical completed wells. However, it has been used in openhole completed horizontal wells for holdup and water velocity measurement, in case of emulsion or in harsh wellbore environment, where the spinners and probes get sticky.

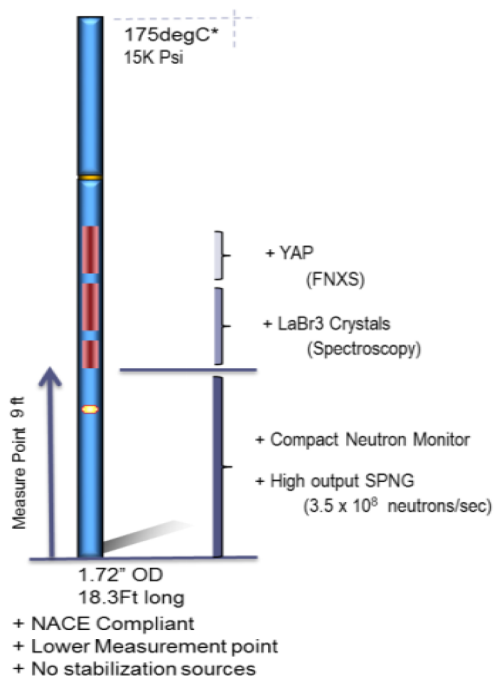


Figure 6—New Generation Pulsed Neutron Tool Design

To fit through most completion restrictions, the tool's outer diameter (OD) is 1.72 in. The housing is corrosion resistant and NACE compliant, allowing deployment in corrosive wellbore environments such as those encountered in the presence of H₂S or CO₂. The tool consists of a pulsed neutron generator (PNG) and four detectors. A compact neutron monitor (CNM), primarily sensitive to fast neutrons, is placed adjacent to the high-output PNG where it can most accurately and precisely measure source output. Three scintillation gamma ray detectors are used. The two detectors nearest to the source (near and far) have LaBr₃ scintillators. The farthest-spaced detector (deep) uses an yttrium aluminum perovskite YAP scintillator.

In this study, the new Pulsed Neutron Logging (PNL) tool was run primarily for water holdup and water velocity analysis, to be integrated with the multi-array Production Logging Tool (PLT) data—particularly in the event of downhole debris or the presence of emulsions.

Job Planning

Pre-job planning is critical in complex wells, particularly those with challenging trajectories, restrictions, and washouts. Before any toolstring design, detailed planning is essential to ensure that the logging objectives can be met safely and effectively.

To address these challenges, a job planning simulation is run before job to verify and ensure that sufficient reach can be achieved. This includes selecting the appropriate number of tractor drives and tool string

components required to navigate the wellbore and reach total depth (TD). The job planner also allows to evaluate tractor force requirements and cable tension during both down and up passes.

This step is vital not only for ensuring successful conveyance but also for identifying potential risks, such as sticking. It enables the Production Engineer to take appropriate precautions, such as designing a contingency plan or integrating release mechanisms, so that in the event of tool sticking, the bottom-hole assembly (BHA) can be safely released and retrieved via fishing operations if needed.

Below is an example of job planning where the number of tractor drives was optimized to confirm adequate force to reach TD and to validate cable tension limits under realistic well conditions (Fig-7).

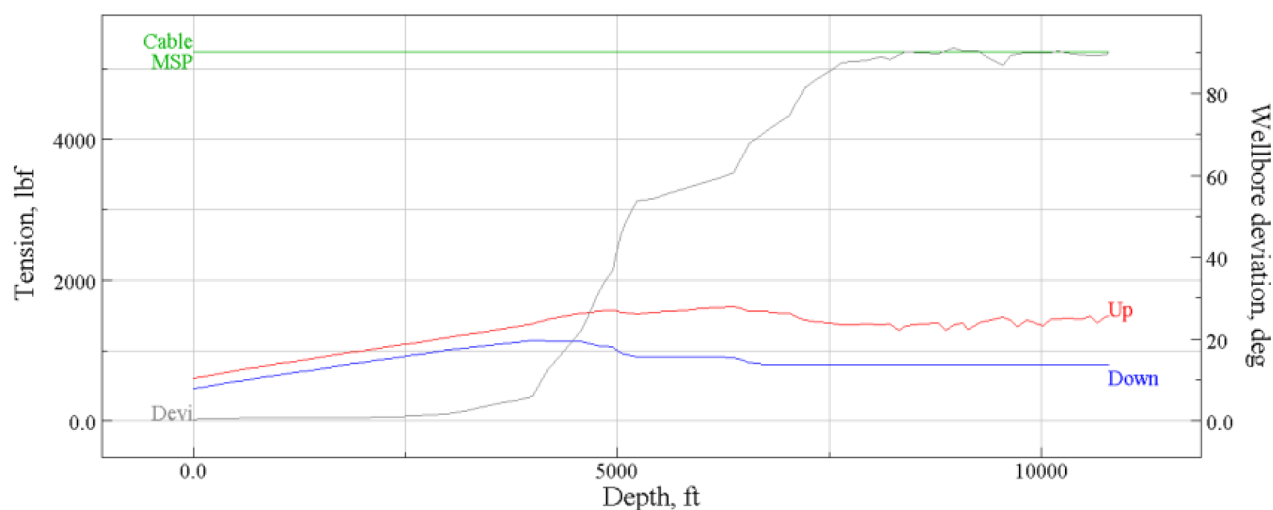


Figure 7—Wireline Tractor Simulation

Logging Results and Discussion

These following case studies focus on a major onshore oil field in the Middle East, the primary producing reservoir is a carbonate sequence from the Late Jurassic period, characterized by interbedded limestone and occasional dolomitic layers. Initial development relied on vertical wells but has since advanced through the adoption of modern technologies such as horizontal drilling and integrated reservoir monitoring systems to enhance recovery and optimize performance.

Well 1 – Tractor Conveyed MPLT in Openhole Completion

The objective of the logging operation was to determine the flow profile across the horizontal 6 $\frac{1}{8}$ " openhole interval and to detect any water entry. The well had a maximum deviation of 90°, and the minimum restriction in the wellbore was 2.635". The total production of the well is high with a 37% water cut.

Logging was performed using a slim openhole tractor under both flowing and shut-in conditions. The successful data acquisition in this high deviation well is a direct result of smart engineering design, careful planning, and the effective deployment of openhole tractor technology. Given the 90° deviation and restricted wellbore geometry, the use of the tractor was critical to overcoming conveyance challenges and enabling full coverage of the horizontal section.

Openhole tractoring in such complex environments demands a high level of operational precision and tool reliability. The toolstring was engineered to minimize risk and maximize depth of investigation, while ensuring stability and consistent tool motion, key factors for accurate flow profiling.

As a best practice, spinner calibration was conducted in the vertical section during both surveys to ensure data integrity. The logging included long, consistent down and up passes across the horizontal section, contributing to the reliability and repeatability of the measurements.

The results and flow profile obtained from the multi-array tool are presented in Figure 8. The downhole flow profile is classified as semi-uniform, with only 76% of the openhole interval contributing to the total flow, indicating that certain zones dominate production. The primary oil-producing zones are Interval A (43% of total oil) and Interval B (28% of total oil). The main water-producing zones are Interval B (35% of total water) and Interval C (39% of total water). No crossflow was observed during either the flowing or shut-in surveys.

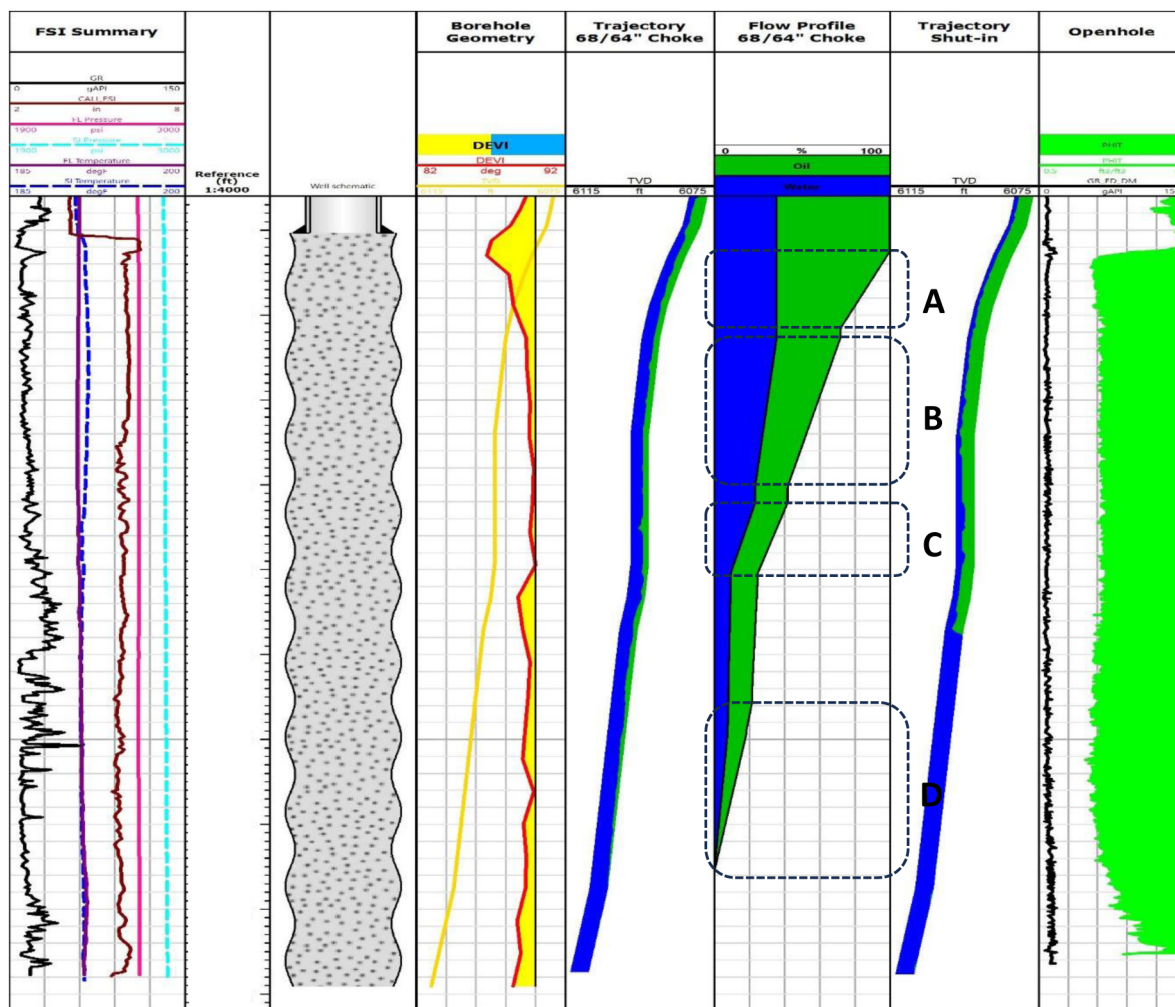


Figure 8—Well 1 – Multi-array Production Logging Results

This high-quality dataset underscores the importance of integrating reliable tractoring, robust logging design, and well-executed field engineering to achieve successful diagnostics in challenging horizontal wells.

Well 2 – Tractor Conveyed MPLT and PNL in Screen Completion

This well was initially converted from an openhole to an Inflow Control Device (ICD) Screen completion based on insights obtained from a previous multi-array logging survey. Although the advanced tractor deployed in this operation was originally designed for openhole applications, it demonstrated exceptional efficiency, automation, and adaptability in a cased hole environment. The tractor successfully navigated through wellbore restrictions, delivering smooth, high-quality data acquisition despite the complex geometry.

This operation showcases the value of smart engineering and tool flexibility, particularly in wells where retrofit completions and restricted access could otherwise compromise data quality. The ability of the tractor to maintain tool stability and consistent motion in a cased hole setting was critical in achieving reliable and repeatable logging passes.

The well was completed with a 4.5" ICD screen completion, comprising a total of 21 screens or inflow points. At the time of logging, the well was high oil with 46% water cut. The logging program consisted of one down and one up pass of MPLT data, one PNL pass, and 12 water flow log stations using PNL. Vertical spinner calibration was also performed to ensure data integrity.

Figure 9 displays the raw data on the left and the interpreted flow profile along with the PNL data on the right. Notably, the PNL was not deployed for traditional reservoir saturation interpretation, but rather to support Flow Spinner Interpretation by providing water velocity and phase holdup information, particularly in the presence of potential downhole emulsions.

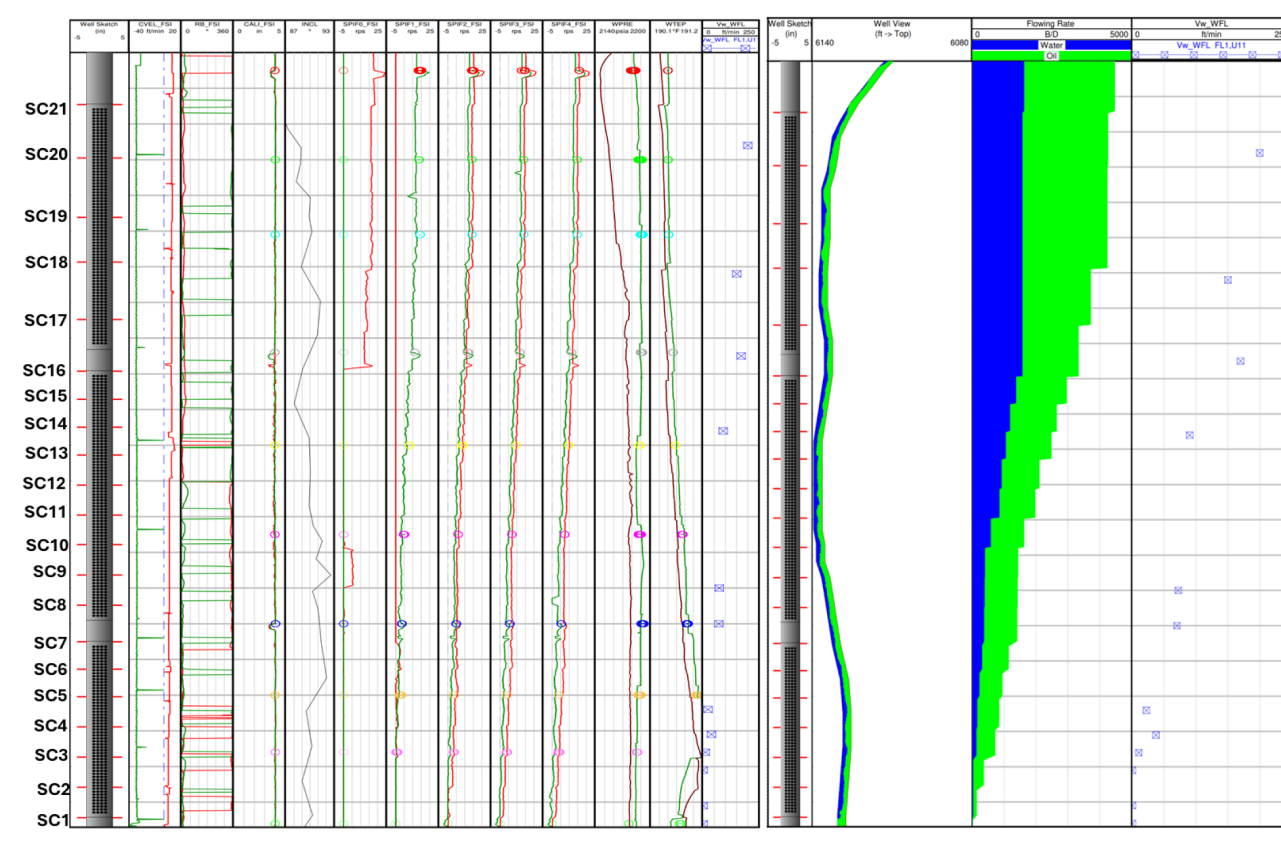


Figure 9—Well 2 – Multi-array Production Logging Data and Results

The results show that the main oil contributors are Screens 17 to 21, while the primary water-producing zones are Screens 8 to 16, accounting for 74% of total water production. Minimal water was observed from Screens 17 to 21. Based on this detailed flow characterization, a decision was made to shut off Screens 8 to 16 to reduce water cut and optimize oil recovery.

This case reinforces the importance of integrating advanced logging tools, versatile tractor technology, and multi-source data fusion (MPLT + PNL) to deliver actionable insights—even in retrofitted and cased horizontal wells.

Conclusions

The deployment of a slim E-line tractor to convey the multi-array production logging tool (MPLT) in both horizontal openhole and cased hole oil producers has proven to be highly effective, delivering the logging objectives with precision and efficiency. The tractor's ability to navigate through high-deviation wells and restricted completions, whether openhole or completed with ICDs, demonstrated excellent reach, stability, and data quality, even in complex well environments.

When compared to traditional coiled tubing (CT) conveyance, the E-line tractor offers several distinct advantages:

- Reduced operational risk and production impact: E-line interventions are less invasive, leading to minimal production or injection loss during the operation, unlike CT, which typically requires more intrusive setups.
- Significant time saving: A conventional CT-based horizontal logging operation can take 5 to 7 days, factoring in mobilization, site preparation, rig-up, and rig-down. In contrast, the E-line tractor operation was completed in just 48 hours, achieving an 60% reduction in operational time.
- Smaller environmental footprint: E-line setups require significantly less surface infrastructure and energy, leading to lower carbon emissions with more than 70% CO₂ emission reduction and reduced HSE exposure.
- Cost effective with cost reduction by more than 50%.
- Conservation of ground water which is usually required for pressure testing and other requirement in CT operations.
- Creating market competition for better service quality.

The comprehensive production profiling across the reservoir enabled a detailed evaluation of the downhole oil and water zonal contributions, which are critical inputs for water shut-off strategies and ICD conversion decisions leading to the isolation of water producing ICD screens based on reliable flow diagnostics.

Moreover, the integration of multiphase flow data with Pulsed Neutron log (PNL) measurements, not for saturation, but to capture water velocity and phase holdup in emulsified flow, enhanced the accuracy of the interpretation. These datasets will be incorporated into the reservoir model to refine the understanding of fluid movement, sweep efficiency, and reservoir behavior over time.

Overall, this approach highlights the operational, economic, environmental, and strategic benefits of combining advanced tractoring technology with smart data acquisition and interpretation workflows, transforming well surveillance into a proactive, high-value activity.

Nomenclature

CALI	: Caliper
CH	: Cased Hole
CT	: Coil Tubing
CNM	: Compact Neutron Monitor
CVEL	: Cable Velocity
Devi	: Wellbore deviation
FNXS	: Fast Neutrons Cross-Section
GR	: Gamma Ray
H₂S	: Hydrogen Sulfide
ID	: Internal diameter
MD	: Measured Depth

MPLT : Multiphase Production Logging Tool
OD : Outside diameter
OH : Open Hole
PLT : Production Logging Tool
PNG : Pulsed neutron generator
PNL : Pulsed Neutron Tool
POOH : Pull-out of Hole
RIH : Running in Hole
SPIF : Array Spinner
TVD : Total Vertical Depth
V_w : Water Velocity from PNL
WPRE : Wellbore pressure
WTEP : Wellbore Temperature
YAP : Yttrium aluminum perovskite YAP

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